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EDUCATION

New York University, Tandon School of Engineering

MS, Computer Science, May 2026. GPA 3.67/4.0

IIIT Delhi

B.Tech, Computer Science and Applied Mathematics; minor in Economics, 2020 - 2024. CGPA 8.19/10

Coursework: Stochastic Processes, Statistical Inference, Convex Optimization, Game Theory, Econometrics, Valuation and Portfolio Management, Machine Learning

PROJECTS

trading-agent (not yet public) | Python, pandas, LightGBM, TabPFN, Chronos-2 2026

- Cost-aware evaluation harness (purged walk-forward, CPCV, Deflated Sharpe, PBO, random-portfolio null) over 14 models on 8 US ETFs, 2011 to 2024.
- No candidate beat vol-targeted equal weight after costs (best 0.68 Sharpe vs 0.81), so none was promoted; zero-shot Kronos called 5-day direction right 46.6%.
- Execution path shipped regardless: Alpaca and Zerodha Kite adapters, a \$100 paper account on launchd, drawdown throttle, kill switch, four gates before real money.

Non-stationary bandits | Python, NumPy, Matplotlib github.com/happyc0der/multi_armed_bandit_algorithms

- Five piecewise-stationary bandit algorithms (TS-GE, AdSwitch, M-UCB, UCB1, epsilon-greedy) behind one interface, benchmarked from 2 to 128 arms.
- Audited TS-GE against its paper and found the Beta update reversed: TS-phase regret 21,860 to 174; mandatory probing is 75.9% of its total regret.
- Made AdSwitch 59x faster (218.6 s to 3.7 s) and cut memory from 2,048 MB to 272 MB at K=128, T=1e6, bit-identical output.

Kaggle S6E9, EV purchases | Python, LightGBM, XGBoost, CatBoost, github.com/happyc0der/kaggle-s6e9-ev-purchases
PyTorch

- Kaggle Playground S6E9 (ROC AUC): out-of-fold 0.94648, public leaderboard 0.94649 against a leader at 0.94674; final ranking pending.
- Nested target encodings over the synthetic generator's artifacts on 668,665 rows, 20 folds; logit blend of LightGBM, CatBoost and a 15-seed MLP.
- 192 experiments judged by paired DeLong tests; label resampling showed a perfect model scores 0.9459 ± 0.0003 , so the gap left is label noise.

Subgradient descent with momentum | Python, PyTorch, NumPy, scikit-learn, Matplotlib github.com/happyc0der/AOMML

- Stochastic subgradient descent and heavy-ball momentum as torch.optim.Optimizer subclasses with constant, $1/\sqrt{k}$ and $1/k$ schedules, on LASSO and a ReLU network.
- 89 tests in CI: autograd matches the closed-form subgradient to $4e-16$, LASSO optimum within $7.4e-5$ of scikit-learn, updates match torch.optim.SGD to $1e-14$.
- Momentum's apparent 9.3e8x speedup falls to 1.05x once the effective step is matched; its real benefit is stability at large steps.

COMPETITIONS

- **Biohub Cell Tracking, Kaggle**, 2026. Public leaderboard 0.952. Final ranking pending, competition ends 2026-09-29.
- **Kaggle Playground S6E9**, 2026. Out-of-fold AUC 0.9465. Final ranking pending, competition ends 2026-09-30.
- **TrafficFlowBench**, 2026. Public leaderboard 0.845 against a 0.553 baseline. In progress, ends 2026-11-06.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash, eZ80 assembly, Monkey C

ML: PyTorch, scikit-learn, LightGBM, XGBoost, CatBoost, Hugging Face transformers, vLLM, Ollama, NumPy, pandas, polars

Data: PySpark, Databricks, SQL, PostgreSQL, SQLite, pandas, Streamlit, Altair, matplotlib

Tools: Git, GitHub Actions, Docker, AWS (CDK, App Runner, DynamoDB), Vercel, Linux, WSL2, Tailscale

ACTIVITIES

- Events head, Finnexia (finance club), IIIT Delhi; organised "The Critical Bid"
- Attended the first International Quantum Communication Conclave, New Delhi, 2023